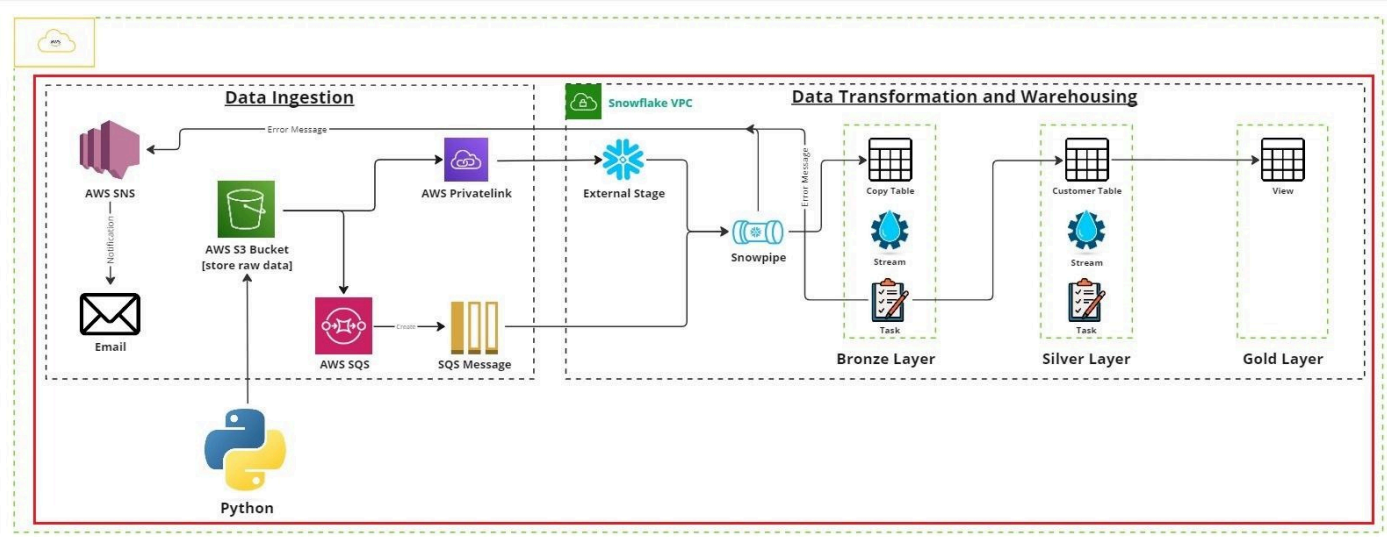
Technical Overview

In this document, we are going to set up snowpipe in snowflake to ingest the source data in the customer source table from the AWS S3 bucket

Architecture



Pre-requisites:

- AWS Account
- Snowflake Account

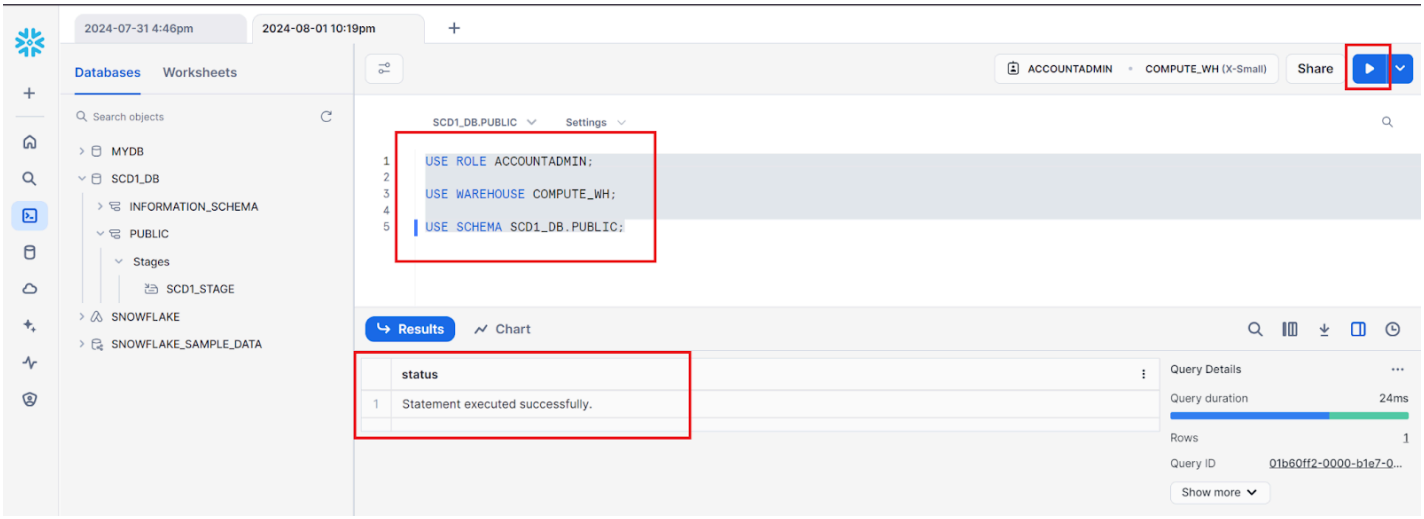
Create a Customer Source table in Snowflake

- Go to the snowflake console and open the already opened SQL worksheet page (if not opened, open a new SQL worksheet) and execute the below SQL statements to select your role,warehouse and schema accordingly by clicking the play button on the top right corner

```
USE ROLE ACCOUNTADMIN;

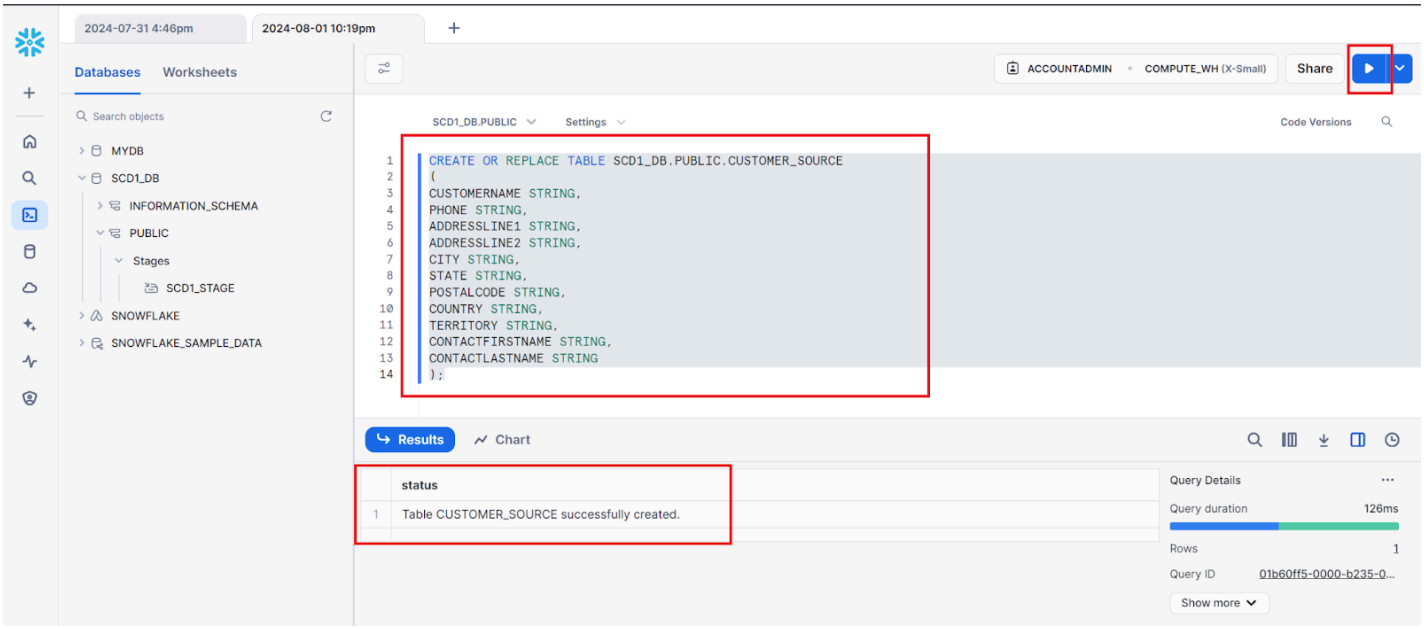
USE WAREHOUSE COMPUTE_WH;

USE SCHEMA SCD1_DB.PUBLIC;
```

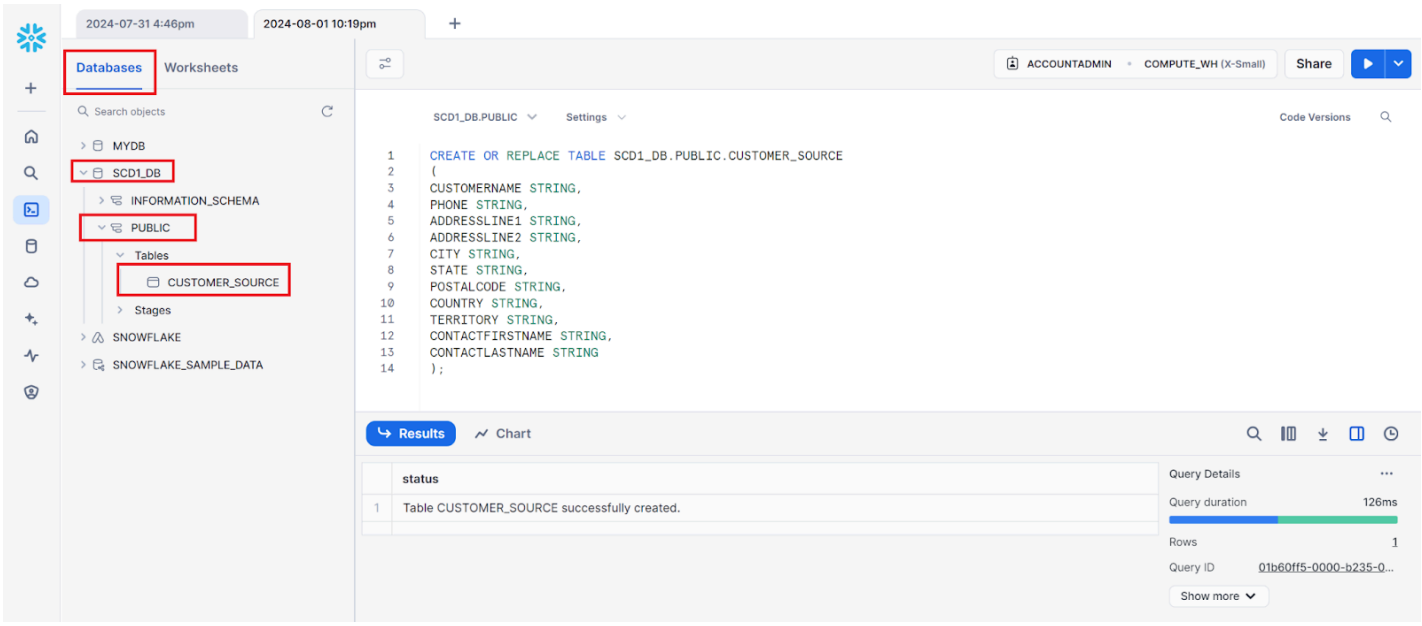


- Execute the below code in the snowflake worksheet to create a table to store the source data from the S3 data bucket through snowpipe

```
CREATE OR REPLACE TABLE SCD1_DB.PUBLIC.CUSTOMER_SOURCE
(
CUSTOMERNAME STRING,
PHONE STRING,
ADDRESSLINE1 STRING,
ADDRESSLINE2 STRING,
CITY STRING,
STATE STRING,
POSTALCODE STRING,
COUNTRY STRING,
TERRITORY STRING,
CONTACTFIRSTNAME STRING,
CONTACTLASTNAME STRING
);
```



- Once the table is created successfully, refresh your page to validate that the table is created under the correct database and schema.



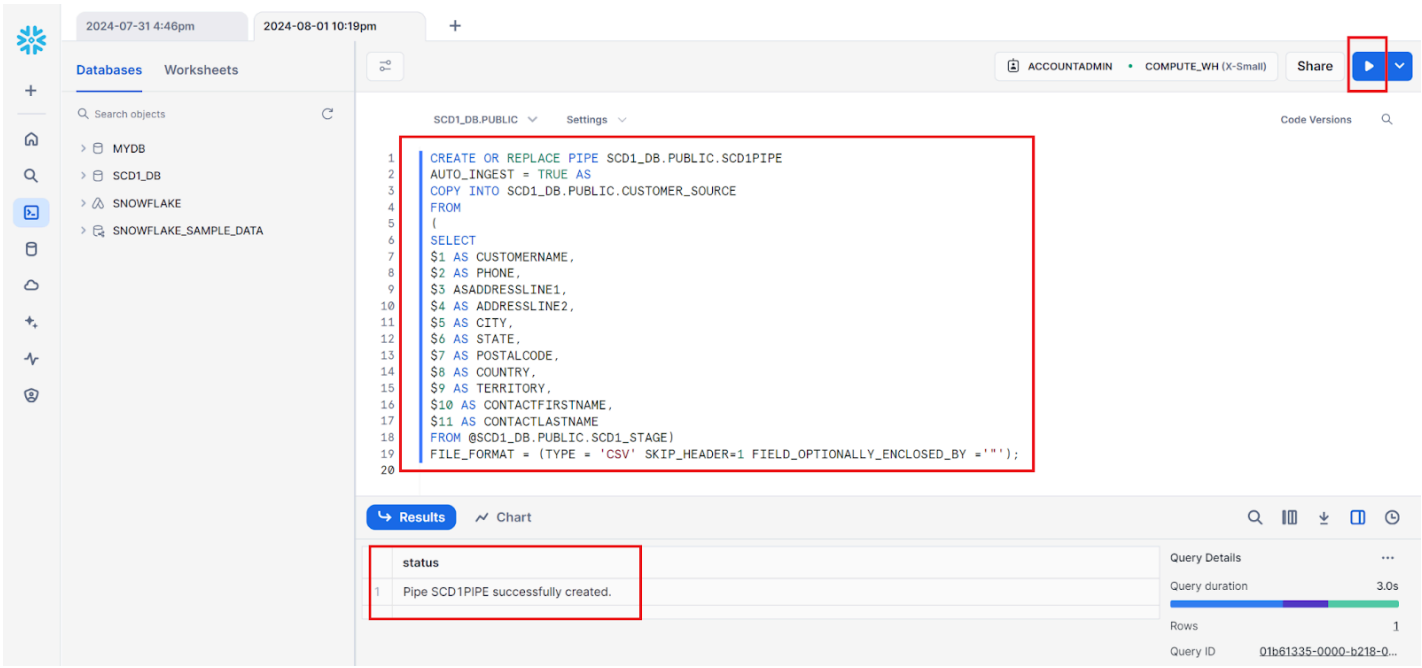
Setup the snowpipe in Snowflake

- Execute the below code in the snowflake worksheet to setup a pipe to load the data from S3 bucket to the customer source table created previously.

CREATE OR REPLACE PIPE SCD1_DB.PUBLIC.SCD1PIPE

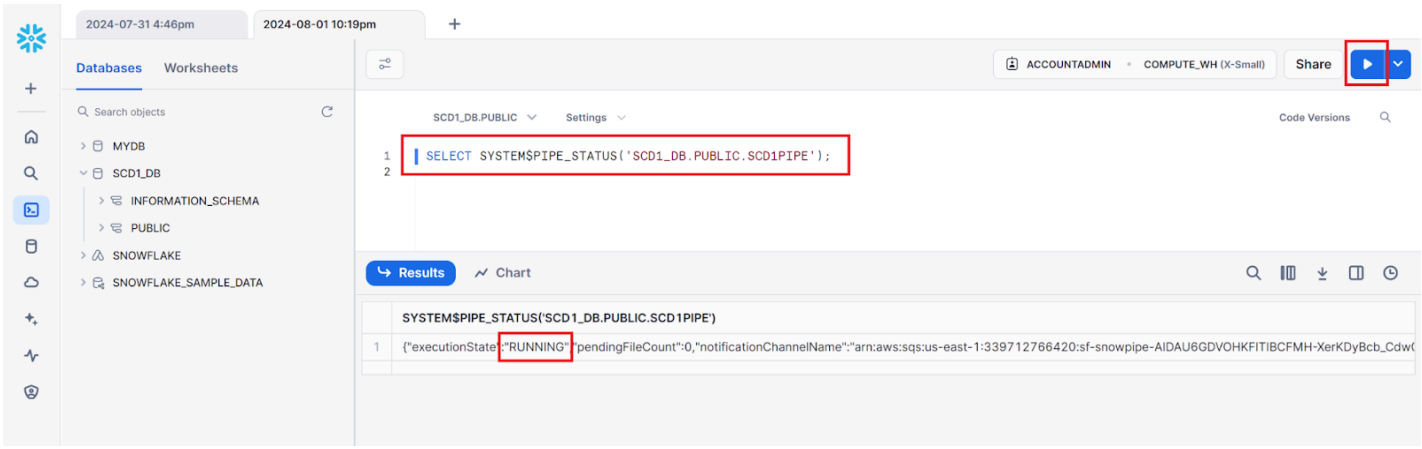
AUTO_INGEST = TRUE AS

```
COPY INTO SCD1_DB.PUBLIC.CUSTOMER_SOURCE
FROM
(
SELECT
$1 AS CUSTOMERNAME,
$2 AS PHONE,
$3 ASADDRESSLINE1,
$4 AS ADDRESSLINE2,
$5 AS CITY,
$6 AS STATE,
$7 AS POSTALCODE,
$8 AS COUNTRY,
$9 AS TERRITORY,
$10 AS CONTACTFIRSTNAME,
$11 AS CONTACTLASTNAME
FROM @SCD1_DB.PUBLIC.SCD1_STAGE)
FILE_FORMAT = (TYPE = 'CSV' SKIP_HEADER=1 FIELD_OPTIONALLY_ENCLOSED_BY ='"');
```



- Verify if the above created snowpipe is in the RUNNING state by running the below SQL query in the snowflake worksheet

SELECT SYSTEM$PIPE_STATUS('SCD1_DB.PUBLIC.SCD1PIPE');

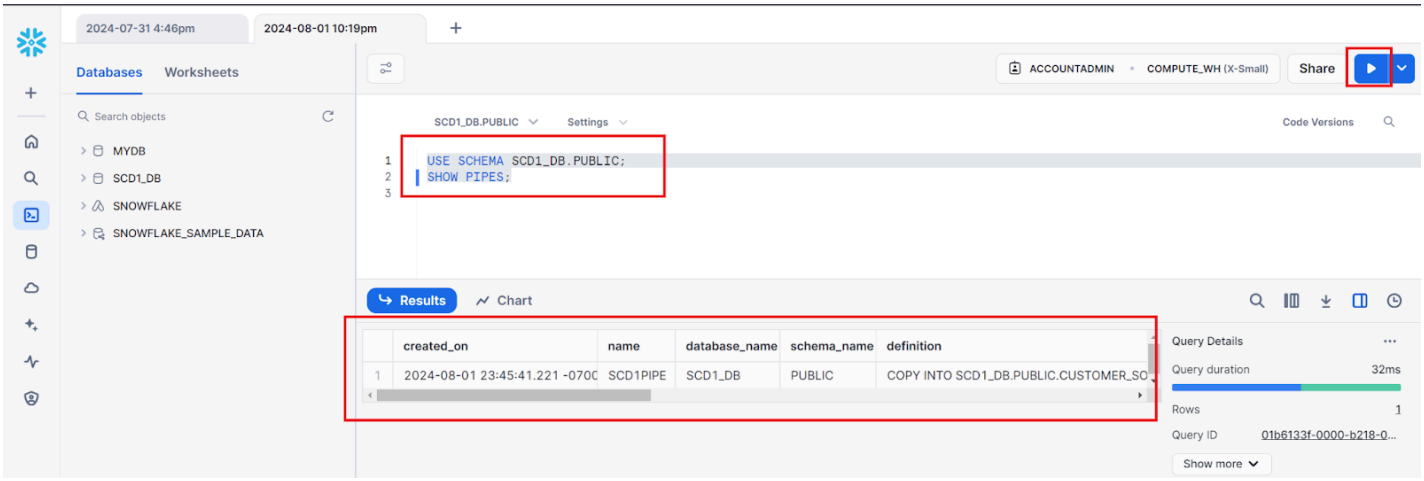


Configure Event Notification for snowpipe

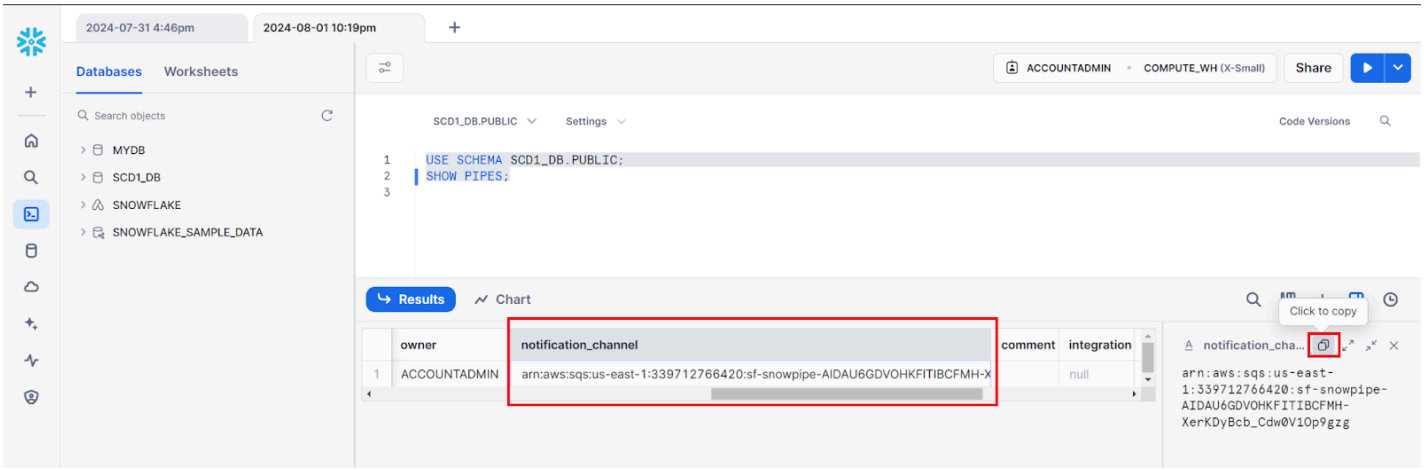
- Run the following code in the Snowflake worksheet to retrieve the notification channel value for the pipe created in the previous step, which is needed to configure the event notification.

USE SCHEMA SCD1_DB.PUBLIC;

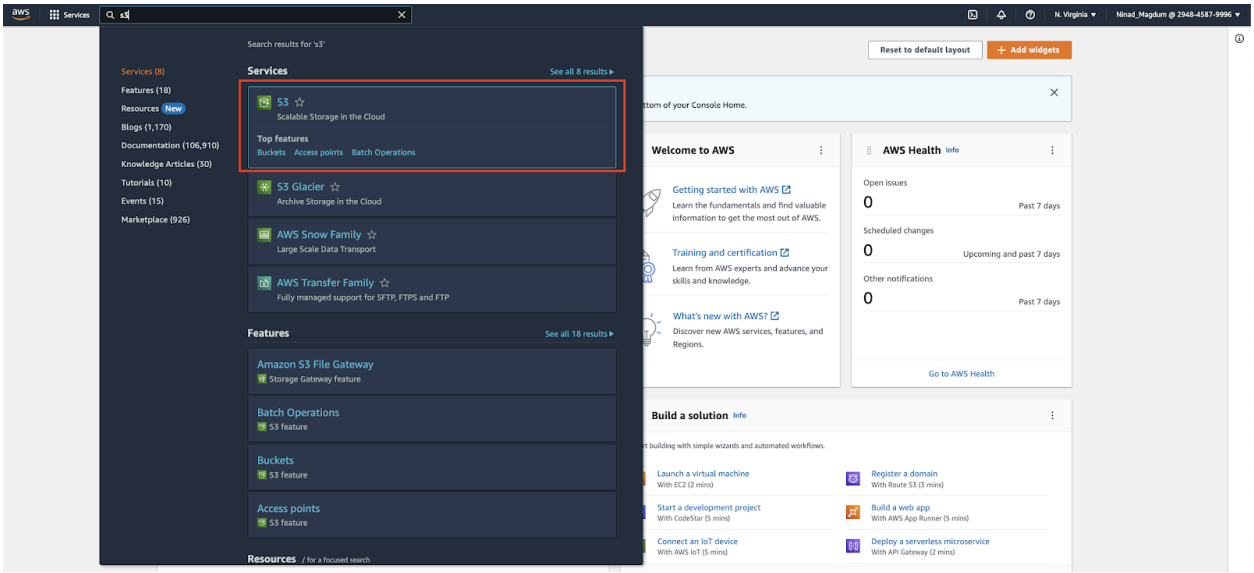
SHOW PIPES;



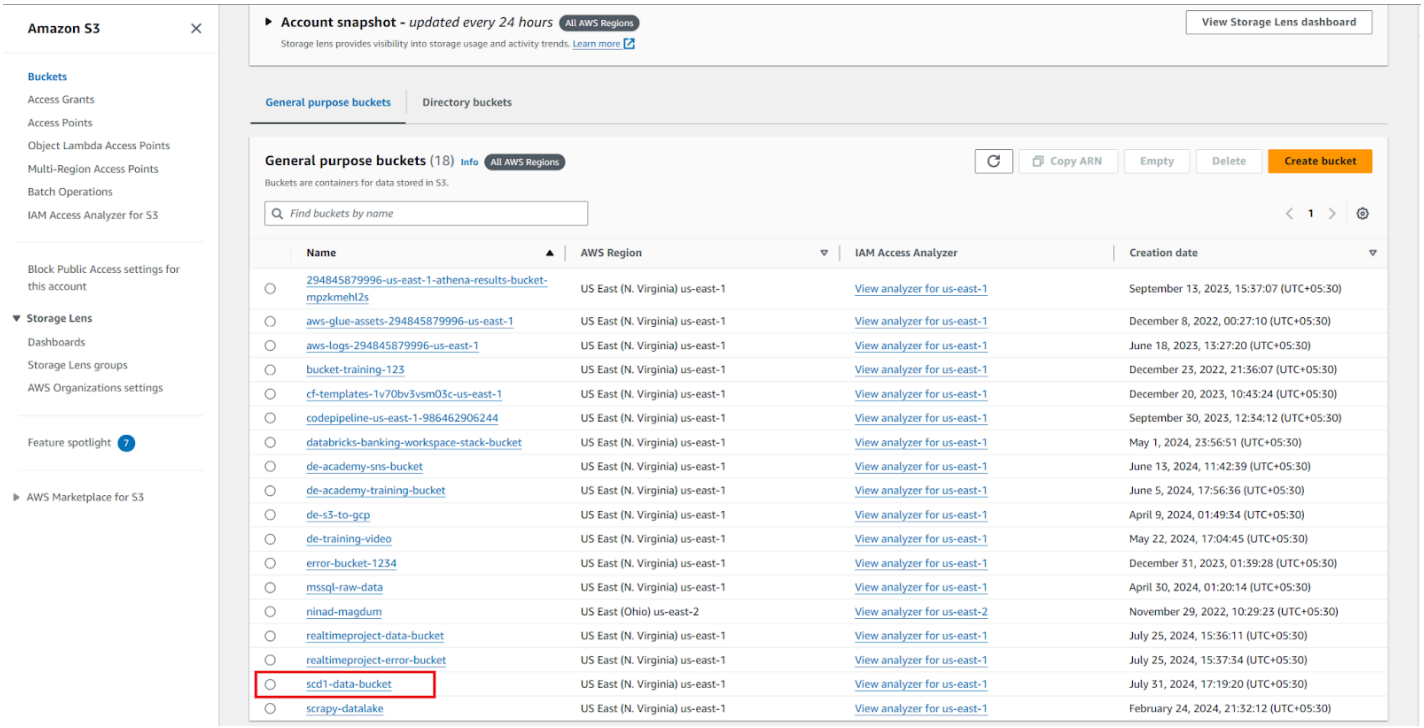
- Copy and note the ARN of the SQS queue in the notification_channel column for the next step



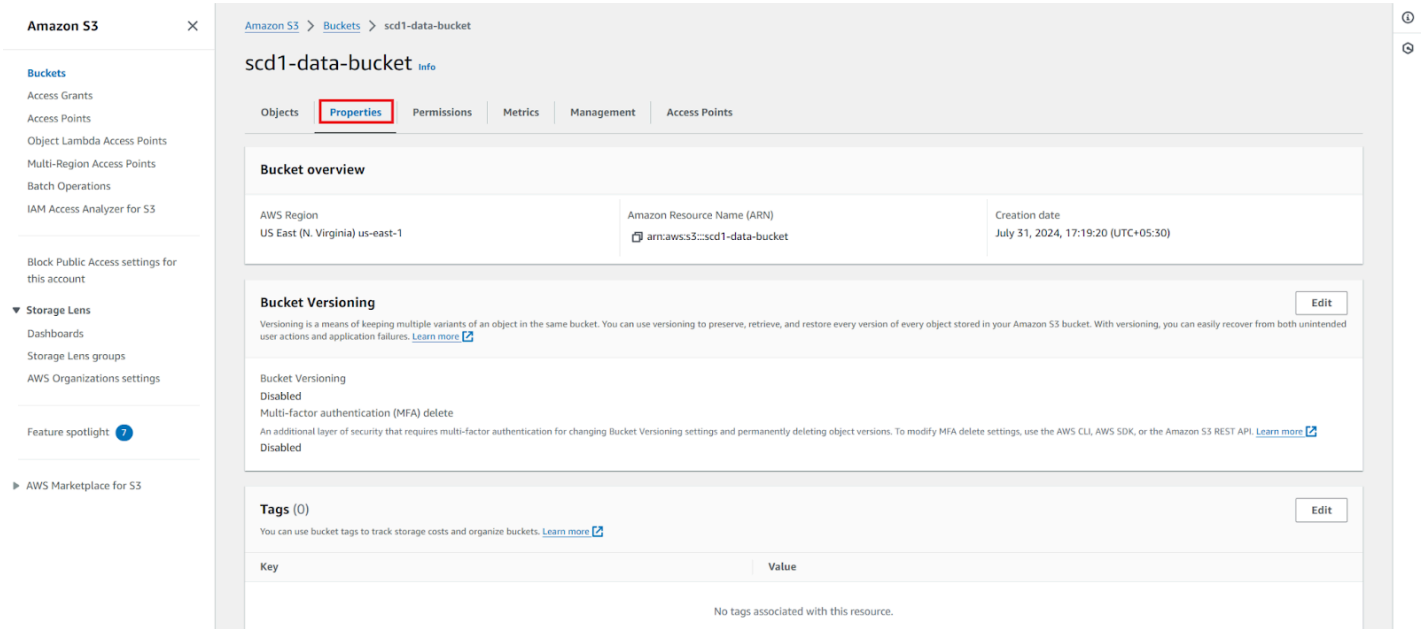
- Go back to your AWS Management Console and find AWS S3 by entering "S3" in the search bar and open it



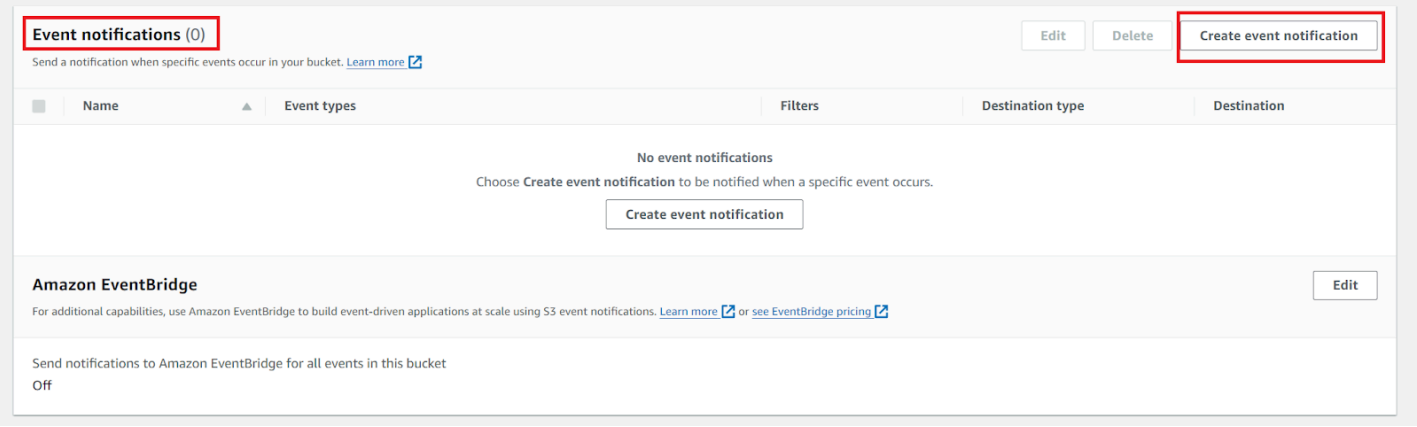
- From the list of buckets find your scd1 data bucket and click it to open



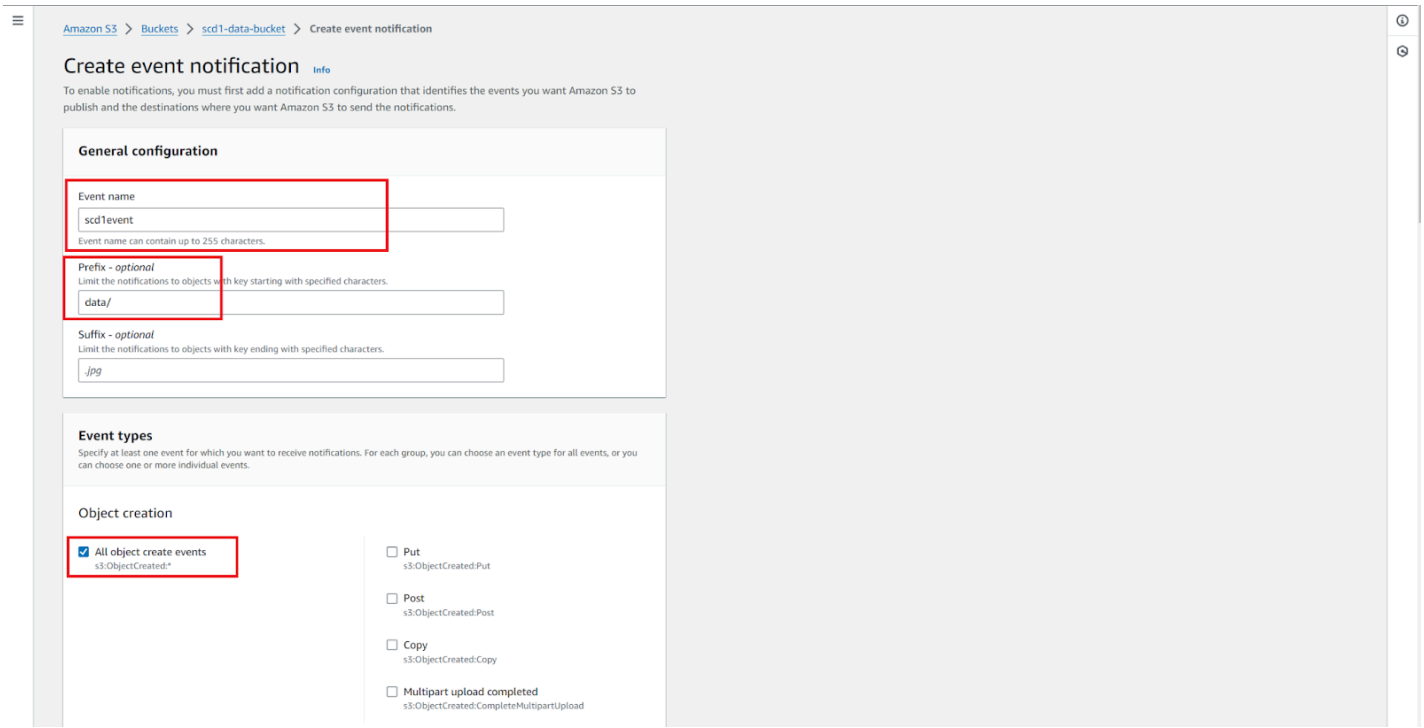
- On the next page click on Properties.



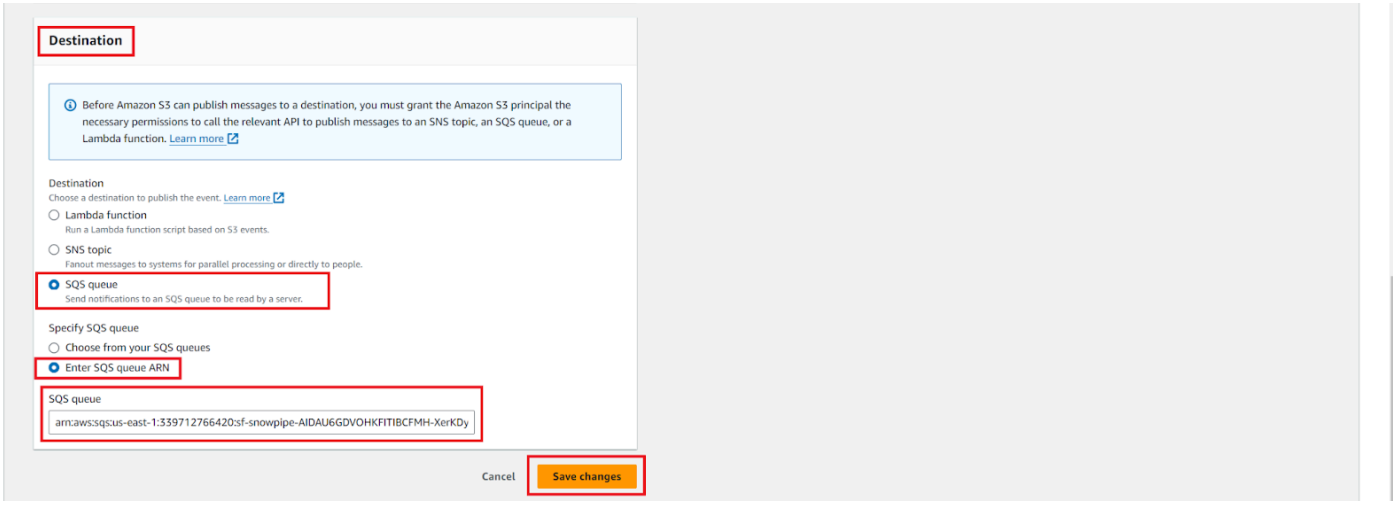
- Scroll down to the Event Notification section and click on Create event notification.



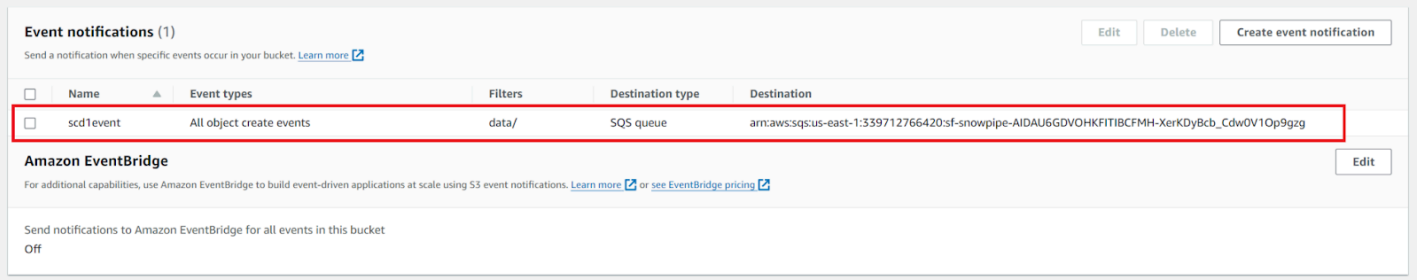
- Give a name to the Event notification, under prefix give the folder name as data/ and select All object create events.



- Now scroll down to the Destination section and select SQS Queue as destination, choose to Enter SQS Queue ARN and paste below the value of the notification_channel column that you copied by doing SHOW PIPES in snowflake in the previous steps. Then click on Save changes.



- Your event notification for your snowpipe is now set up successfully under the Event notification section of S3



VOILA! YOUR SUCCESSFULLY CREATED SNOWPIPE AND AUTO INGEST HAS BEEN ENABLED

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